# Instructions for Students – Preliminary Design Report (PDR), Critical Design Report (CDR) and NOC Submission

All participating students are requested to follow the instructions below while submitting their PDR and CDR for SAVe 2.0 and ASV 2026:

1. **Registration:** All participants must register using the registration link provided by the organizers.

2. **No Objection Certificate (NOC):** Students must obtain a **No Objection Certificate (NOC)** from the competent authority of their respective institution. Submission of the duly signed and stamped NOC, in the prescribed format, is **mandatory** for eligibility to participate in the competition. Failure to submit the NOC will result in the application being **summarily rejected**. The NOC template is provided on the subsequent pages of this document (Refer page No 7-8).

4. **File size and Format:** The PDR must be submitted only in PDF format and maximum of 3 MB.

5. **Faculty Mentor:** Every team must have a Faculty Mentor. Teams without a Faculty Mentor will not be considered eligible.

6. **Official Communication:** All official communications regarding the competition will be made only through the Faculty Mentor and Team Leader.

7. **PDR/CDR Submission:** The PDR/CDR must be submitted via email to: **save.niot@gov.in** and **auvniot@gmail.com.** A separate report needs to be submitted.

8. **File Naming Convention:** The PDF file shall be named using the following format:

   Example: `NSCP_2026_REG_0001_PDR/CDR.pdf`

9. **Submission Deadline:** The PDR and CDR must be submitted as per time line given on NSCOT website. Late submissions will not be considered.

10. **Verification:** Students are advised to ensure that all required documents are complete, correctly named, and submitted before the deadline. Incomplete or incorrectly formatted submissions may be rejected.

10. **Important Note**

Participants are advised to verify that:

* All required documents are complete.

* The PDR/CDR is in PDF format.

* The file name follows the prescribed naming convention.

* The PDR/CDR is submitted before the deadline.

* The Faculty Mentor's details are correctly provided.

11. Contact Details:
R Sundar
Programme Coordinator – NSCOT
National Institute of Ocean Technology
Ministry of Earth Sciences, Govt of India
Chennai 600100
save.niot@gov.in |auvniot@gmail.com

## Preliminary Design Report (PDR) Format

The Preliminary Design Report (PDR) should be prepared in the following order:

1.Title Page

- Project Title
- Team Registration ID
- Institute Name
- Student Name(s)
- Faculty Mentor Name

2.Abstract

- A concise summary of the proposed Autonomous Underwater Vehicle (AUV), including the objectives, methodology, and expected outcomes.

3. Introduction

- Background of the project
- Problem statement
- Objectives and scope of the proposed work

4. Literature Review

- Review of relevant research, existing AUV technologies, and related work.
- Identification of the research gap and motivation for the proposed design.

5. Theoretical Design

- Overall system architecture
- Mechanical design
- Electrical and electronic systems
- Sensors and actuators
- Control strategy
- Software architecture
- Design methodology, calculations, and expected performance

6. Experimental Results

- Preliminary simulations, calculations, prototype development, or experimental validation (if available).
- Discussion of the obtained results.

7. Acknowledgements

- Acknowledge the contributions of the institute, Faculty Mentor, laboratories, funding agencies (if any), and individuals who supported the work.

8. References

- List all references cited in the report using a standard citation style (e.g., IEEE, APA, or another accepted academic format).

Note:

- The report should be 5 to 10 pages in length.
- Submit the report only in PDF format.
- Include clear figures, tables, and diagrams wherever necessary.
- Ensure that all content is original and properly referenced to avoid plagiarism.

## Critical Design Report (CDR) Format

This CDR format maintains continuity with the PDR submission guidelines while emphasizing design maturity, engineering validation, manufacturability, testing readiness, and project execution, which are the primary objectives of a Critical Design Review. A separate report needs to be submitted. The CDR shall be prepared in the following order wherever applicable .

1. Title Page
    - Project Title
    - Team Registration ID
    - Institute Name
    - Competition Category
    - Student Name(s)
    - Faculty Mentor Name
    - Date of Submission

2. Executive Summary
    - Project objective
    - Mission requirements
    - Design overview
    - Key innovations, if any.
    - Current development status

3. Introduction
    - Background
    - Problem statement
    - Objectives
    - Scope of work
    - Design requirements

4. Design Evolution

Describe the improvements made after the Preliminary Design Review.

- Review comments received during PDR
- Design modifications
- Updated specifications
- Engineering justification for changes

5. System Requirements
    - Functional requirements
    - Performance requirements
    - Operational requirements
    - Environmental constraints
    - Mission profile

6. Overall System Architecture
    - System block diagram
    - Functional architecture

- Mechanical architecture
- Electrical architecture
- Software architecture
- Communication architecture

7. Mechanical Design
   - CAD model
   - Structural design
   - Material selection
   - Dimensions
   - Weight estimation
   - Buoyancy and stability analysis with numerical calculations
   - Waterproof enclosure design
   - Manufacturing methodology

8. Electrical and Electronics Design
   - Power distribution system
   - Battery specifications
   - Circuit diagrams
   - Electronic modules
   - Wiring diagram
   - Protection mechanisms

9. Sensors and Actuators
   - Sensors
   - Thrusters/Actuators
   - Navigation sensors
   - Communication modules
   - Interface specifications

10. Control and Software Architecture
    - Software flowchart
    - Control algorithms
    - Navigation strategy
    - Mission planning
    - Embedded software architecture
    - Communication protocols

11. Simulation and Analysis
    - Mechanical analysis
    - Structural analysis
    - CFD analysis (if available)
    - Control simulations
    - MATLAB/Simulink results
    - Performance predictions

12. Prototype Development Status
    - Fabrication status
    - Assembly status
    - Photographs of prototype
    - Manufacturing progress
    - Procurement status

13. Testing and Validation Plan
    - Component testing
    - Integration testing
    - Functional testing
    - Water trials (if applicable)
    - Performance validation
    - Acceptance criteria

14. Safety and Risk Assessment
    - Hazard identification
    - Risk analysis
    - Safety mechanisms
    - Emergency shutdown procedure
    - Failure recovery strategy

15. Project Management
    - Work Breakdown Structure (WBS)
    - Team responsibilities
    - Project schedule
    - Gantt chart
    - Remaining milestones
    - Risk mitigation plan

16. Bill of Materials (BoM)
    - Component list
    - Quantity
    - Estimated cost
    - Vendor details (if available)
    - Total project cost

17. Conclusion
    - Current design readiness
    - Expected outcomes
    - Remaining activities before final competition

18. Acknowledgements
    - Institute

- Faculty Mentor
- Sponsors/Funding Agencies (if any)
- Technical mentors

19. References

List all references using a standard citation format such as IEEE or APA.

20. Appendices

- CAD drawings
- Datasheets
- Circuit diagrams
- Engineering calculations
- Test reports
- Additional photographs

Important Notes

- The report shall be 10–20 pages in length.
- Submit only one PDF file containing the complete report and appendices.
- Include clear figures, tables, photographs, and diagrams wherever necessary.
- All engineering calculations shall be properly explained.
- All content shall be original and appropriately referenced to avoid plagiarism.
- The CDR should demonstrate that the design is sufficiently mature for fabrication, integration, and testing prior to the final competition.

**(Institute Letter Head)**

**NO OBJECTION CERTIFICATE (NOC)**

**Date:** ___________

**To Whomsoever It May Concern**

This is to certify that the following student(s) of ________________________________________________ **(Name of the Institution)** are bonafide student(s) of our institution and are hereby permitted to participate in the National Student Challenge Programme in Ocean Technology (NSCOT) **2026 Competition** organized by the **National Institute of Ocean Technology (NIOT), Chennai**, under the Ministry of Earth Sciences, Government of India.

**Team Details**

- **Team Name:** ______________________________________
- **Registration No:** _____________________________________

| Sl. No. | Student Name | Roll No. | Programme/Year | Email ID of Students |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |

**Faculty Mentor**

- **Name:** ___________________________________________
- **Designation:** _____________________________________
- **Department:** _____________________________________
- **Email:** ___________________________________________
- **Mobile:** __________________________________________

The institution has **no objection** to the above-mentioned student(s) participating in the competition and developing the proposed autonomous marine vehicle (SAVe 2.0/ASV) as per the competition guidelines. The students will participate under the guidance of the above-mentioned Faculty Mentor.

The institution confirms that:

1. The above student(s) are bonafide students of this institution.
2. The Faculty Mentor has been nominated by the institution to guide the team.
3. The institution has no objection to the students participating in the competition and related activities.
4. The students and Faculty Mentor shall abide by the rules, regulations, and decisions of the competition organizers.
5. The institution understands and accepts the terms and conditions of the competition.

This certificate is issued upon the request of the student(s) for the purpose of participating in the NSCOT 2026 Competition.

**Authorized Signatory**

Signature: ___________________________

Name: _______________________________

Designation: _________________________

Institution: __________________________

Official Seal of the Institution